REPLICATION STUDY · MOEX 2015–2020 · HSE PRACTICUM · JULY 2026

Statistical arbitrage that earned Sharpe 2.6 in Warsaw goes flat in Moscow

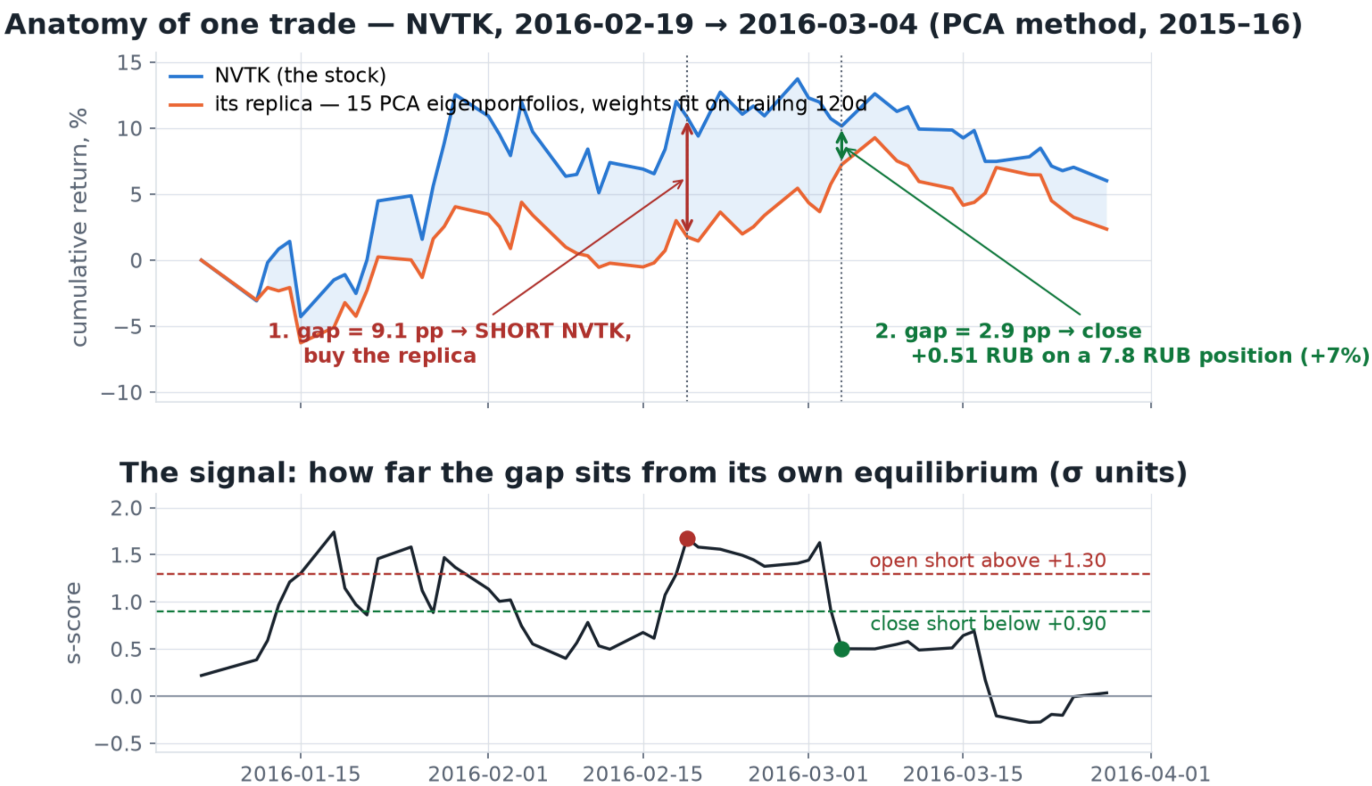
The Avellaneda–Lee factor/Ornstein–Uhlenbeck strategy from Adamczyk & Dąbrowski's Polish study (arXiv 2512.03073), re-implemented end-to-end and re-run on the 60 most liquid Russian stocks with identical windows, filters, thresholds and costs. **Verdict: the pipeline transfers, the alpha does not.** Over 2017–2019 the gross (pre-fee) edge is exactly zero, every method finishes at or below the RUB cash baseline, and the paper's ranking of methods fully inverts. The one profitable period, 2015–2016, turns out to be a single quarter of post-crisis dispersion rather than an edge.

Universe 60 MOEX names (top MOEXBMI weights ≈ WIG20+mWIG40) · **Methods** PCA eigenportfolios · index-ETF factors · 8 sector TR indices · per-stock LSTM · **Risk-free** 1Y OFZ: 7.34% (2017–19), 10.39% (2015–16), 4.86% (2020) · **Costs** 0.1%/side · **Leverage** 2:1 · W=120 · κ>4

GROSS EDGE · PCA 2017–19	NET OF CASH · PCA 2017–19	LSTM · 2017–19	Q1 2015 · 111 TRADES
–0.1 ₽	**–26.7 ₽**	**–37.4 ₽**	**+101 ₽**
pre-fee P&L of 1,092 trades per 100 ₽ — the signal is flat, not negative	in Poland the same method made +20% at Sharpe 2.63	the paper's novel method (+10% in Poland) transfers worst of all	more than the entire 2015–16 profit; the other 563 trades lost money

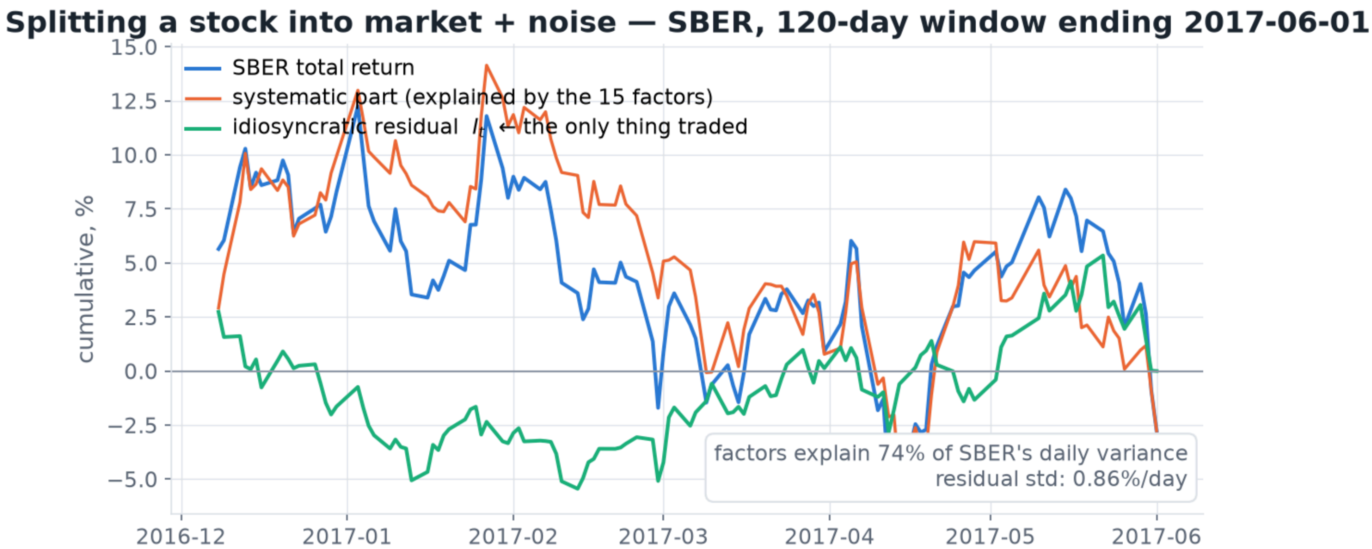
1 · How the strategy works

Classic pairs trading longs asset A and shorts a correlated asset B. This strategy replaces B with a **synthetic replica of A** built from tradable systematic factors (a factor/APT regression). The regression residual is the idiosyncratic component; if it mean-reverts fast enough (κ > 4), the residual's distance from its Ornstein–Uhlenbeck equilibrium — the **s-score** — becomes the trading signal. The three methods differ *only* in how the replica is built; everything downstream is identical.



> One real trade from the Russian backtest. NVTK had run ~9pp ahead of its replica (15 PCA eigenportfolios, weights refit daily on a trailing 120-day window); the s-score crossed +1.30, so the book shorted NVTK and bought the replica in equal RUB size, closing 10 days later once the gap had narrowed to ~3pp: +0.51 RUB on a 7.8 RUB position. The market's direction cancels out — only the gap matters.

2 · Where the tradable signal comes from



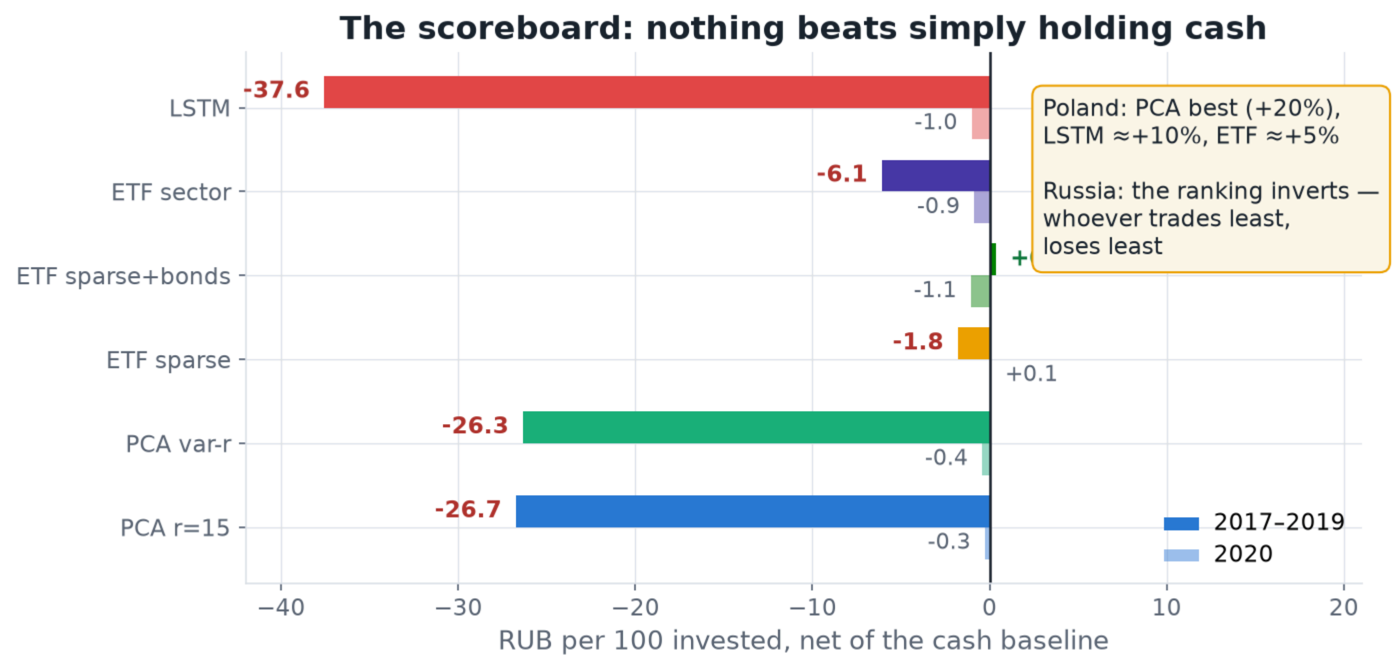
> Each stock's return is split into a systematic part (explained by the factors — 74% of SBER's daily variance) and an idiosyncratic residual. Only the residual is traded. It is a bridge: by construction of the regression it returns to zero at the window's end, and the bet is that when it wanders far from equilibrium mid-window it comes back — fast enough to beat costs.

> **A common suspicion, ruled out first:** the Russian market is *not* too concentrated for this decomposition. Prior-year correlation matrices need **13–16 principal components** to explain 55% of variance (λ_1 = 13–24%) — essentially the same dimensionality the paper reports for Poland (16–18). The APT machinery ports fine; what is missing is the payoff.

3 · The result: everything converges to cash minus fees

Every method with the paper's own Polish-optimized thresholds, on Russian data. The dashed line is the honest benchmark: the paper's accounting compounds idle cash at the risk-free rate, and RUB cash paid 7.3% over 2017–19 (vs 1.5% in Poland). Everything that looks like a gain is cash, not alpha — the ETF curves merely track the baseline while PCA and LSTM fall below it. In 2020 nothing blew up and nothing profited: all ten configurations land within ±1.4 RUB of cash.

4 · The scoreboard



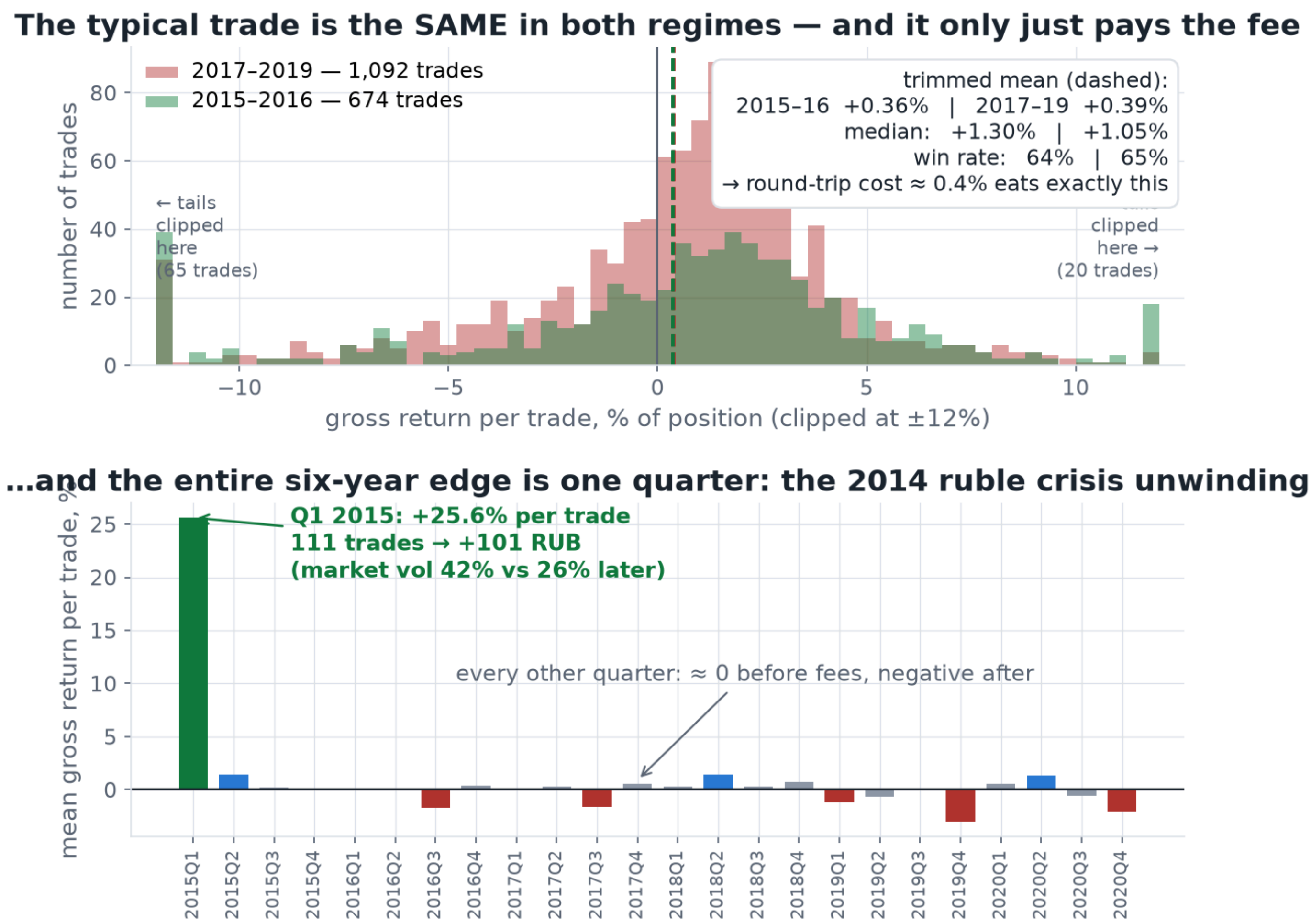
> Final equity minus the cash baseline, per method and period. Positive means the strategy added value over doing nothing. Only one configuration is positive, by 0.4 RUB — a rounding error.

METHOD	RU TOTAL	RU VS CASH, ₽	RU SHARPE 17/18/19	TRADES	POLAND (PAPER)
PCA r=15	−2.1%	−26.7	−1.08 / +0.88 / −2.66	677	+20% · 2.63/1.01/−1.16
PCA var-r	−1.7%	−26.3	−1.08 / +0.88 / −2.28	675	+20% · 2.51/0.44/−0.91
LSTM	−12.9%	−37.4	−0.20 / −0.60 / −2.27	711	≈+10% · 0.60/2.09/−1.53
ETF sparse	+22.9%	−1.8	−0.86 / +0.23 / +0.28	188	+5% · −0.25/−0.46/+1.43
ETF sparse+bonds	+25.0%	+0.4	−0.86 / +0.60 / +0.96	194	—
ETF sector	+18.6%	−6.1	−0.88 / −0.38 / −0.41	276	≈+5%

α = final equity − cash baseline, RUB per 100 invested, 2017–2019, paper's thresholds. Re-optimizing the thresholds on Russian 2015–16 data (the paper's own procedure) does not rescue it: PCA still ends −16.4 ₽ out-of-sample.

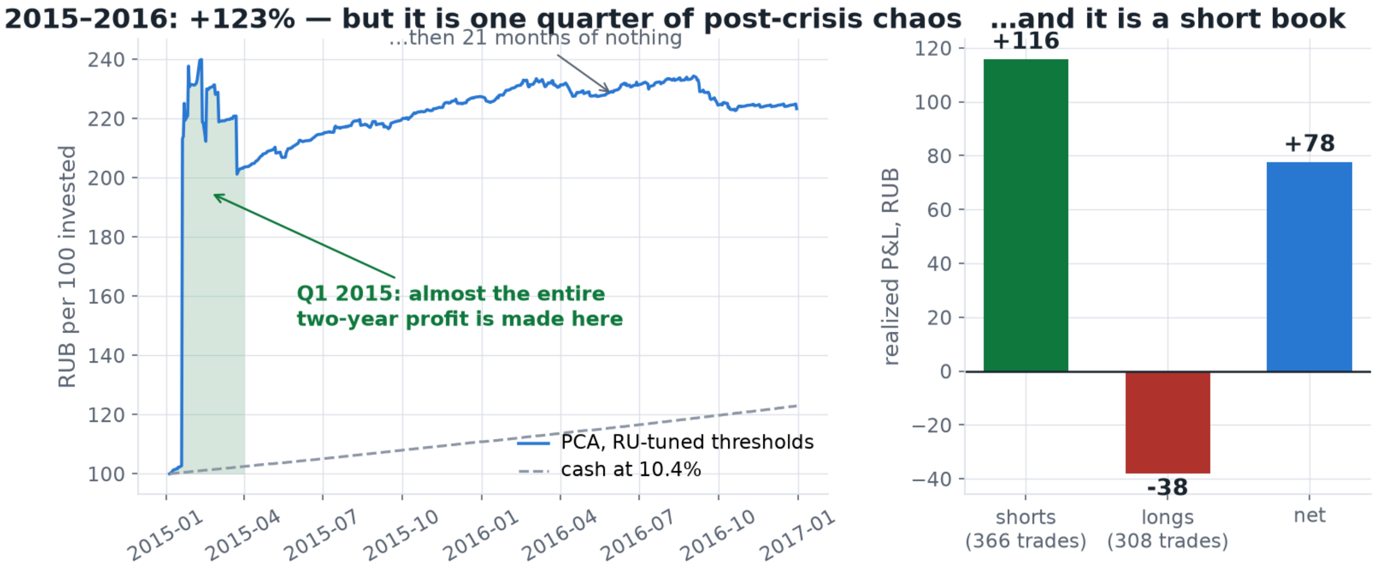
5 · What 2015–2016 really was: a tail, not an edge

Re-tuned on Russian data, PCA earns **+123%** across 2015–2016 against a 23% cash baseline. Going to trade level shows what that number actually is.



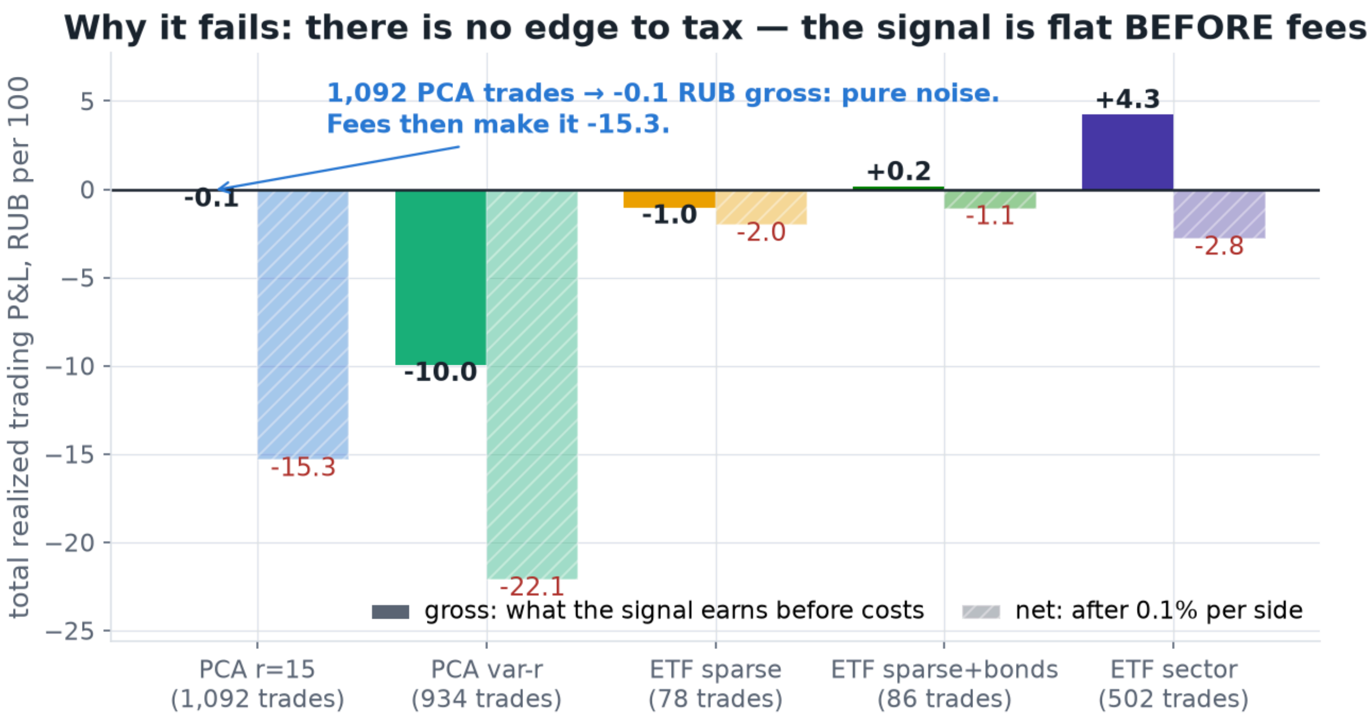
Top: the distribution of gross-of-fee returns per trade. The typical trade is statistically indistinguishable between the two regimes — trimmed mean +0.36% vs +0.39%, median +1.30% vs +1.05%, win rate 64% vs 65% — and in both it earns almost exactly the ~0.4% round-trip cost. Bottom: mean gross return per trade by quarter. The entire six-year edge is Q1 2015.

The strategy never had a steady edge in Russia: **its median trade merely pays the toll booth.** The whole 2015–16 profit is one fat right tail — **Q1 2015's 111 trades made +101 ₽ (mean +25.6% per trade), while the other 563 trades lost −4.6 ₽**. That quarter is not a leverage artifact (gross exposure Σ|β| median 2.75, same as the rest of the period) and not a single-name artifact (56 stocks contributed, top-3 = 28%). It is the post-ruble-crisis unwind: market volatility **42%** in Q1 2015 versus **26%** in 2017–2019, and ~80% of it is short-side. On MOEX this is a bet on crisis dispersion — and it paid exactly once in six years.



> The same fact at portfolio level: almost the entire two-year equity gain lands in Q1 2015, then 21 months of nothing — and the profit is a short book (+116 vs −38 RUB). Mid-cap borrow in 2015 Russia was scarce and expensive, and no borrow fee is modelled, so a good part of the short leg was not realizable in practice.

6 · Why it fails in 2017–2019



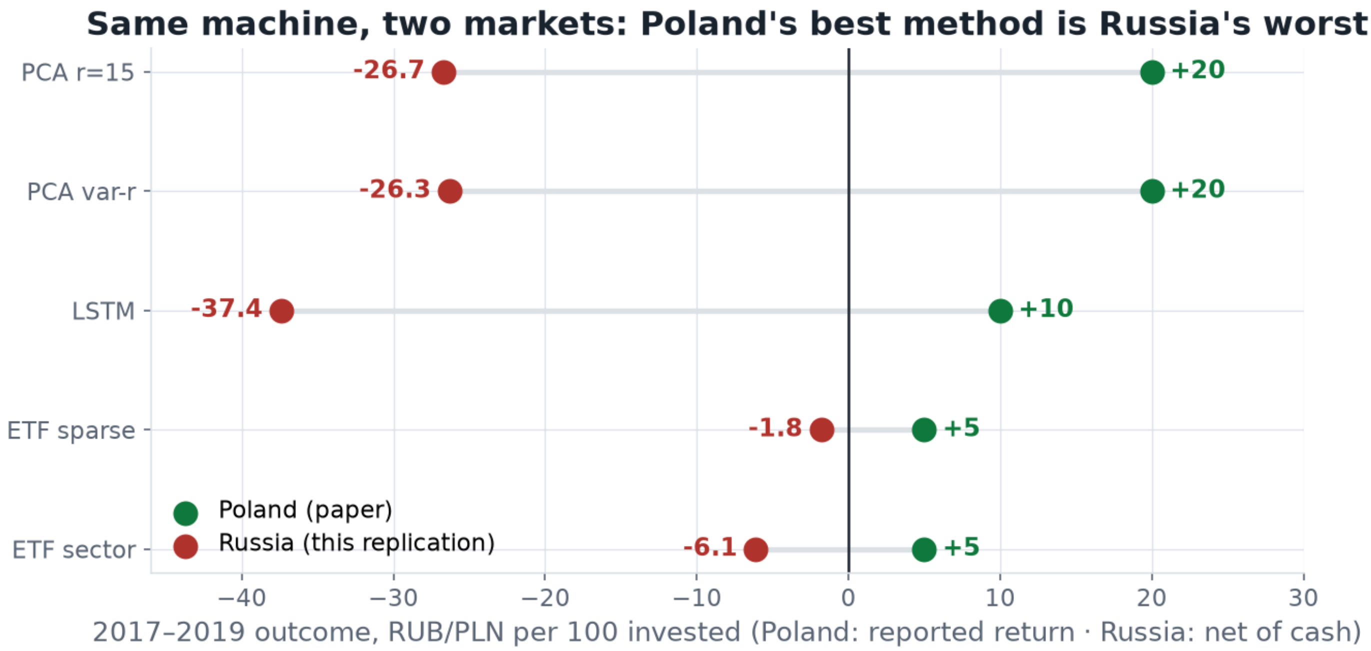
> 2017–2019 with each method's RU-tuned thresholds — the configuration most favourable to the strategy. If Russia merely had expensive trading, the solid (gross) bars would be clearly positive; they hover at zero, and fees then convert ≈nothing into a loss.

- **Not the factor structure.** 13–16 PCs for 55% of variance ≈ Poland's 16–18.
- **The residual pays nothing.** Gross-of-fee P&L 2017–19: −0.1 ₽ from 1,092 PCA trades; +4.3 ₽ from 502 sector-ETF trades. A zero-mean signal before costs.
- **Fees finish it.** ~0.4% per round trip × 500–1,100 trades = 7–19 ₽ per period.
- **Exit asymmetry.** Win rates look healthy (58–62%) but the average win is only 0.5–0.6× the average loss: fast same-side exits cap winners while losers ride to the far threshold.
- **Cash at 7–10% raises the bar.** A hedged RUB book must clear a money-market leg that the paper's accounting hands out free at Poland's 1.5%. Even the best in-sample sparse-ETF configuration (2015–16,

+18.6%) lost to cash (+23.0%).

- **And the cost model still flatters it.** Fees are charged on the *net* replica notional $|1+q^M| \approx 2$, but the replica is traded leg by leg and its *gross* book $\Sigma|\beta|$ runs median 3.0 (p95 5.5, max 32) for PCA — real costs are ~2–3× what the formula charges. Every α here is therefore optimistic.

7 · Russia versus Poland



> Poland's ranking runs PCA > LSTM > ETF; Russia's runs exactly backwards — the methods that trade least (few index factors, few signals) simply lose least. A ranking that inverts across markets suggests the original ordering reflected the Warsaw regime rather than a property of the method.

Conclusions

- **The pipeline replicates; the alpha does not.** Every mechanical element — factor construction, OU fit, s-scores, threshold logic, accounting — ports to MOEX without modification and behaves correctly (validated on synthetic OU data where the model is true by construction, and where the same engine earns +170%). What Russia 2017–2019 lacks is the raw material: residual mean reversion worth more than a round trip.
- **It is a crisis-dispersion bet, not an income stream.** The typical Russian trade earns ≈+0.4% gross in *both* regimes — exactly its own transaction cost — so expected value is zero plus a tail. That tail arrived once, in Q1 2015, and never again.
- **The paper's cross-method ranking is market-specific**, and its "ETFs are robust in crashes" conclusion degenerates in Russia's 2020 to "everything is cash".
- **This reframes the Polish result too:** 2017–19 Warsaw may simply have been a dispersion-rich regime. That is directly testable by re-running the original code on later GPW data — the natural next study.

Caveats

Survivorship: fixed 60-name universe requiring full 2014–2020 history (the paper does the same); delistings such as URKA, MFON and DIXY are excluded — this *flatters* results, so the negative verdict is conservative. **Shorting:** no borrow fees modelled (nor in the paper), which matters for the short-heavy 2015–16 P&L. **Investability:** sector and SMID total-return indices were not directly tradable before 2018 (the paper splices ETF and index history the same way; FXRL tracked MCFTR from March 2016). **Data:** MOEX candles are not dividend-adjusted, so dividends were merged from MOEX ISS and dohod.ru per-payment records and applied on T+2 ex-dates, with share-basis rescaling for post-window splits (GMKN and TRNFP 1:100, PLZL 1:10, VTBR 5000:1); QIWI's pre-2016 payments are missing. **Metrics:** Sharpe uses the paper's realized-only equity convention; a marked-to-market variant tells the same story. **Rates:** period-average 1Y OFZ zero-coupon yields, where the paper used constants.

Appendix · Data, universe and reproduction

Sources. MOEX ISS — daily candles (TQBR board), historical index compositions (MOEXBMI), the zero-coupon government yield curve (`zcyc`) and dividend records; dohod.ru — per-payment dividend history before 2018, where ISS is incomplete. No API keys and no paid vendors: every number in this report is reproducible from public endpoints.

Two traps worth knowing if you rebuild this on MOEX data: ISS candles are *retroactively split-adjusted* while dividends are recorded as paid per the old share, so dividends must be rescaled by the candle/as-traded price ratio before adjusting (GMKN and TRNFP 1:100, PLZL 1:10, VTBR 5000:1); and the paper's per-trade P&L formula is written in *gross* returns — reading it as net charges every trade a phantom unit of notional, which a synthetic-data harness catches immediately.

Universe (60 stocks)

Top MOEXBMI constituents by average index weight across yearly snapshots 2015–2020, requiring ≥95% trading-day coverage from March 2014: AFKS, AFLT, AKRN, ALRS, BANE, BANEP, BSPB, CHMF, FEES, FESH, GAZP, GCHE, GMKN, HYDR, IRAO, IRGZ, KBTK, LKOH, LNTA, LSRG, MAGN, MGNT, MOEX, MRKC, MRKP, MSNG, MSRS, MTLR, MTLRP, MTSS, MVID, NKNC, NKNCP, NLMK, NMTP, NVTK, OGKB, PHOR, PIKK, PLZL, POLY, QIWI, RASP, ROSN, RSTI, RTKM, RTKMP, SBER, SBERP, SNGS, SNGSP, SVAV, TATN, TATNP, TGKA, TRMK, TRNFP, VSMO, VTBR, YNDX.

Reproducing from zero

```
python3 moex_data.py        # data: ~20 min of MOEX ISS + dohod.ru requests, cached
python3 test_synthetic.py   # engine sanity: synthetic OU market where the model is TRUE
python3 run_experiments.py  # transfer test + 2015–16 threshold grid + reruns -> results/
python3 run_lstm.py         # LSTM: ~4 h, 240 stock-year models, cached and resumable
python3 make_figures.py     # figures/*.png
```

Full write-up: `RESULTS.md` · step-by-step recreation: `replication_steps.ipynb` · adaptation decisions: `RU_ADAPTATION.md` · method spec (the paper distilled): `REPLICATION_SPEC.md` . Figures generated by `make_figures.py` , this document by `make_pdf.py` . Source paper: Marek Adamczyk & Michał Dąbrowski, *Statistical Arbitrage in Polish Equities Market Using Deep Learning Techniques* (arXiv 2512.03073), itself a Warsaw re-implementation of Avellaneda & Lee, *Statistical Arbitrage in the U.S. Equities Market* (2008).